

MISS LENHART COULDN'T BE HERE TODAY, SO SHE ASKED ME TO SUBSTITUTE.

MATH **MR. MUNROE**

I'VE PUT OUT YOUR TESTS. PLEASE GET STARTED.

MR. MUNROE, MISS LENHART NEVER TAUGHT US THIS.

THAT'S BECAUSE MISS LENHART DOESN'T UNDERSTAND HOW IMPORTANT CERTAIN KINDS OF MATH ARE.

BUT THIS JUST LOOKS --

THIS MATERIAL IS MORE VITAL THAN ANYTHING YOU'VE EVER LEARNED

BUT --

NO BUTS.

— THIS IS A MATTER OF LIFE AND DEATH.

Name: _____

- The velociraptor spots you 40 meters away and attacks, accelerating at 4 m/s^2 up to its top speed of 25 m/s . When it spots you, you begin to flee, quickly reaching your top speed of 6 m/s . How far can you get before you're caught and devoured?
- You are at the center of a 20m equilateral triangle with a raptor at each corner. The top raptor has a wounded leg and is limited to a top speed of 10 m/s .

(Not to scale)

The raptors will run toward you. At what angle should you run to maximize the time you stay alive?
- Raptors can open doors, but they are slowed by them. Using the floor plan on the next page, plot a route through the building, assuming raptors take 5 minutes to open the first door and halve the time for each subsequent door. Remember, raptors run at 10 m/s and they do not know fear.

You MAY turn the page to the next page, but ONLY the next page, before the exam officially starts at 5:30. The next page contains the exam instructions.

MIDTERM EXAM

ECON 2010 – Macroeconomics

Spring 2018

To be completed in-class on Wednesday, March 7, 2018. You will have 2.5 hours to complete the exam (5:30-8:00pm).

Name:

Midterm exam instructions:

1. There are four questions on the exam, each with multiple parts. The questions are about economics, not about Velociraptors.
 2. Read each problem carefully. Just because it looks like a problem you've seen before doesn't mean that the exam asks you to do exactly the same thing in the problem that you've done before.
 3. Pay particular attention to when a question asks you to explain your answer in words, and when you do not need to.
 4. Write your final answers as legibly as possible in pen. Make sure that it is clear to me what your final answer is.
 5. You may use the back of the exam sheets for scratch paper, or request scratch paper from the exam administrator if you need more. If you end up writing a final answer on scratch paper, be very clear in directing me to your answer, and staple scratch paper on which you've written final answers to your exam.
 6. Show your work. Even if your answer is not exactly correct, you will receive credit for setting up the problem correctly. The better that I understand what you've done and why, the more likely you are to receive partial credit.
 7. You may not use a calculator on the exam.
 8. You do not need to reduce fractions for your final answers just for the sake of reducing them, but you may need to reduce some fractions to make comparisons asked for in the problem.
 9. You may not use notes, the textbook, any other materials on the exam, or confer with anyone about the exam while you're taking it. To say it differently, this is a "closed book" exam.
 10. Please do not ask the exam facilitator questions about what you're supposed to do to answer a given question, or what I intended when I wrote a question. Instead, if you think that a question is ambiguous, make a choice in how you interpret the question so that you can answer the question, and *explain your interpretive choice in your answer*. I will take into account your interpretation (if I agree that there is ambiguity) when grading.
 11. Take a deep breath, remember how much you've learned so far, and do your best!
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DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO.

Problem 1 — Thinking About Unemployment Insurance Like an Economist (14 points)

Below is text (in italics) in which an economist discusses thinking through the consequences of an increase in government unemployment insurance. Some of the sentences end with a capital letter in parentheses. To complete the problem, **match none, one, or more than one** capital letter(s) from a given sentence to the principle of economic thinking that it illustrates from the list following the text. **Some sentences could apply to multiple principles; some may apply to none.** If no letters match to a principle, leave the principle **blank** or write “**none**” next to it.

I recommend turning the page to see which principles you can choose from, and then reading the text all the way through once *before* you begin matching sentences to principles. Focus on what each of the sentences *actually says*; just because a sentence mentions “economics” does **not** mean it should be linked to all of the principles of economic thought. Focus on the explicit content.

“Economists like myself are oftentimes trying to think through what happens when government policy changes. One of the reasons is to provide policymakers with better information about what steps they can take to achieve their goals. I don’t tell policymakers what their goals should be, but the right information is a powerful tool regardless of their goals (A).”

In considering the effect of unemployment insurance on the unemployment rate, I usually assume that money is in people’s best interest, and that they’re trying to earn more of it — so, if they have a choice between two jobs, and the only difference between them is the wage, they’ll take the job with the higher wage (B).

So, consider a labor market with just one individual and a bunch of different employers (C). If unemployment insurance pays very little, less than even the lowest paying jobs in the market, if that one individual loses their job, we expect them to quickly take a job that’s willing to employ them. Even if there are potentially better jobs out there, it’s costly to keep searching for a job relative to taking the worse job that pays better than unemployment insurance (D).

Increasing how much unemployment insurance pays out allows individuals to spend longer searching for higher paying jobs. But even if people are getting matched to better jobs, the unemployment rate might be higher as a consequence. Individuals might be willing to quit their jobs — become unemployed — in order to look for better work.

How much the unemployment rate responds to a one unit change in the value of unemployment insurance is the type of number that economists can provide to policymakers (E). Of course, it’s also very important to bear in mind that a lot of other things can change at the same time that a change in unemployment insurance occurs, but we don’t want those to cloud our analysis (F).

In thinking through the problem abstractly, economists can isolate changes in policy from changes in the broader economy, but in the real world, if lots of things change at once, the effects we predict may not be what we actually observe. This is one reason that I think economists have developed a bad reputation; but prediction, especially in the macroeconomy, is really difficult, and more difficult than many people realize.”

- (a) Thinking on the margin (**2 points**)
- (b) Real-nominal principle (**2 points**)
- (c) Simplifying assumption (**2 points**)
- (d) Distinguishing normative from positive thinking (**2 points**)
- (e) Holding other factors/all else fixed (*ceteris parabis*) (**2 points**)
- (f) Rational actors (actors respond to incentives) (**2 points**)
- (g) Equilibrium (**2 points**)
- (h) Opportunity cost (**2 points**)

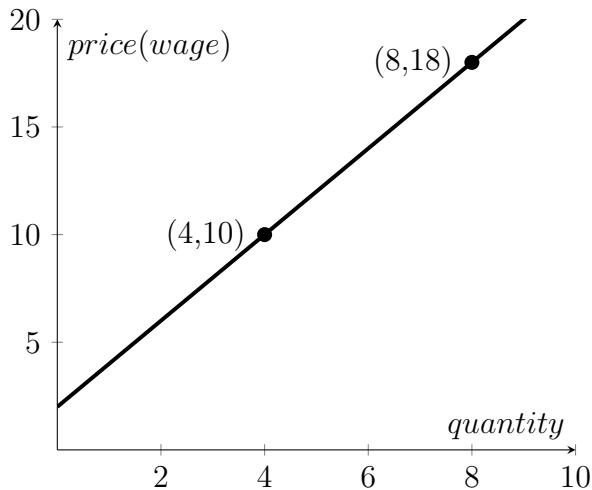
Problem 2 — The Puzzle of Unemployment (44 points)

In this problem you'll discover that there is a puzzle about why unemployment exists *at all* in an economy. To make this discovery, we will consider a simple economy on the fictional island nation of Paradise in which there is only one industry — coconut harvesting. We'll specifically consider the labor market in coconut harvesting. There are 10 coconut harvesters in Paradise, and 2 firms that the 10 harvesters can potentially work for. In Figure 1, I've drawn for you the labor market supply curve for coconut harvesters, the labor market demand curve for coconut harvesters, and a plot that puts both together to find the equilibrium in this labor market.

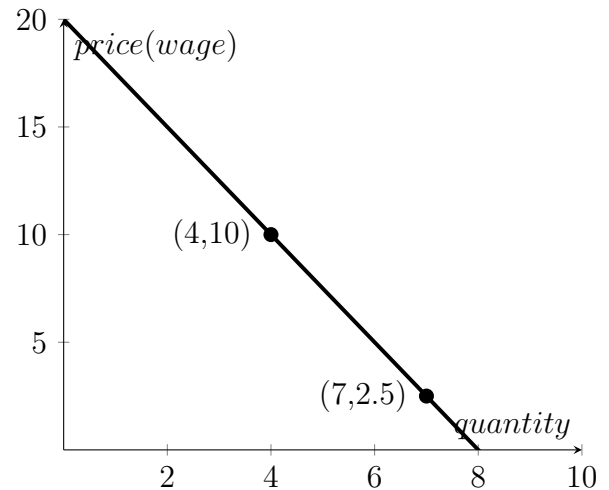
(An aside: You might be interested to note, though it will **not** be important for helping you to solve this problem, that this puzzle of unemployment is fundamentally related to *structural* unemployment, which is different from *frictional* unemployment. Recall from class that I mentioned that your textbook does not do a good job of distinguishing these two types of unemployment from each other, but that they are indeed distinct. This problem will start to uncover the distinction.)

Figure 1: Market supply, market demand and equilibrium for coconut harvesters

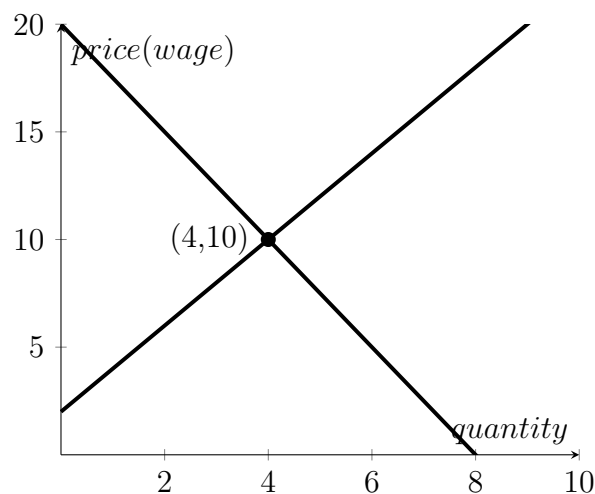
Plot A: Market supply curve



Plot B: Market demand curve



Plot C: Market supply and market demand, a.k.a. market equilibrium



IMPORTANT. Take note of a few measurements and assumptions that you will need to analyze this economy.

- The price (wage) in this labor market is measured as an *hourly wage*.
- The quantity is measured in *hundreds of total hours worked per week* (that means that a 1 on the x axis indicates one *hundred* total hours of work; a 2 on the x axis indicates two *hundred* total hours of work, and a 10 on the x axis indicates 1,000 total hours of work, etc.).
- Assume that each laborer can supply at minimum 0 hours of work, and at most 100 hours per week.
- Assume that all workers are 16+ years old.
- Assume that anyone who loses their job in Paradise looks for work (i.e. remains in the labor force).

- (a) Consider Plot A in Figure 1 above. **True or false:** One of the following *could* potentially explain why an increase in the wage from \$10-18 per hour increases the total quantity of labor supplied in the market from 400 to 800 hours per week: Other firms could open that offer alternative employment to coconut harvesting, Paradise's government increases the cost of training required to be a coconut harvester, and/or the discovery of a reef near Paradise and the opportunity for beautiful scuba diving increases the value of leisure time. In just a few sentences at most, **explain briefly why the statement is true or false** and **mention any relevant principles of economic thinking** that apply to your answer. **(5 points)**
- (b) Consider Plot B in Figure 1 above. **True or false:** It must be the case, given the market demand curve, that at the wage \$2.50, each of the two firms in the coconut market demands exactly 350 (i.e. $\frac{700}{2}$) total hours of coconut harvesting labor. In just a few sentences at most, **explain briefly why the statement is true or false**. **(5 points)**
- (c) What is the equilibrium in the coconut harvesting labor market indicated in Plot C of Figure 1. **(2 points)**

(d) If all 10 workers in Paradise are employed in the coconut harvesting labor market, and they each work an equal number of hours, how many hours, in equilibrium, according to Plot C does each worker work? **(2 points)**

(e) Consider the equilibrium in the coconut harvesting labor market that you found (a.k.a. characterized/described) in part (c) above. Analyze what happens to the market when a new coconut harvesting robot is introduced to the market. Assume that the robots pick coconuts faster than human workers so that it's cost effective for both of the coconut harvesting firms to purchase at least a few of these robots. Moreover, the robot does not help workers pick coconuts faster or better, it is a substitute for their labor. Hold other factors fixed (*ceteris parabis*) as you perform your analysis. **(8 points total)**

(1) Which curve(s) shift as a result of this change? You could say either labor supply, labor demand, none, or both. **(2 points)**

(2) For each curve that you indicated would shift in part (1) above, indicate which direction the curve will shift (**either right or left**). (If no curves will shift, just say "no curves will shift.") **(2 points)**

(3) **Draw** a new market equilibrium plot that contains the curves in Plot C and also reflects the curve shifts that you think would occur according to parts (e1) and (e2) above. **Draw** the previous equilibrium curves in solid lines, and the shifted curve(s) in dashed lines. (If no curves shift, just draw the starting solid curves from Plot C.) **State** how the wage and quantity of labor demanded have changed relative to the equilibrium indicated in Plot C — are they higher, lower, the same, or unknown as compared to before?

Note that I do **not** give you enough information to put specific numbers on the new equilibrium or the changes. Also, you do **not** need to draw the starting equilibrium as precisely as in Plot C (this is not a drawing test!); only well enough for you to show the shifts in the curves you may make, and analyze the equilibrium change. **(5 points)**

- (f) Can the equilibrium change that you found in part (e) above above occurred **without** a change in the unemployment rate? Yes or no? (If you did not find an equilibrium change, just state “no change.”) In just a few sentences at most, **explain briefly why or why not**. Hold other factors besides the shift in curves you found fixed for this analysis (*ceteris parabis*). **(4 points)**

- (g) Consider the plot that you drew in part (e3) above, with the following change. Imagine now that while the quantity of labor demanded can change freely, the wage rate is stuck at the equilibrium level in Plot C — that is, the wage *cannot* fall below \$10 per hour.

Re-draw the full plot that you drew with both original and any shifted curve(s) from (e3) here. **Label** the starting equilibrium (at the point where the initial curves cross) with an **A**. Next, **label** the point that results from both any shifted curve(s) *and* from the stuck wage with a **B**. On the quantity axis, mark the quantity of labor demanded in the starting equilibrium and **label** it Q_A . On the quantity axis, mark the new quantity of labor demanded that

results from any shifted curve(s) and the stuck wage and **label** it Q_B . On the price/wage axis of your plot, mark the wage for equilibrium A and **label** it W^A . On the price/wage axis of your plot, mark the wage for point B and **label** it W^B . If you think some of your points have two labels, that's fine. If you did not draw shifted curves, just re-draw your (e3) plot from above. **(5 points)**

(h) The difference between **which two points** that you labeled in your plot for part (g) above indicates a market with excess labor supply? **(2 points)**

(i) If every worker in Paradise has an individual contract that entitles them to 40 hours per week of work at \$10 or more per hour *if* they are employed, how does the unemployment rate change as a result of the movement from equilibrium A to B in part (g) above? Does the unemployment rate go **up, down, or stay the same**? Assume that the firms honor these contracts, and that there are no other changes in the economy that happen at the same

time (hold all else fixed). In just a few sentences, **explain briefly your answer. (5 points)**

- (j) Can you think of a government policy that we've discussed in class (we considered the effects of this policy at length in one in-class exercise) that might produce a similar effect to the one you discovered in part (i) above where contracts between workers and firms keep wages fixed at \$10 per hour? **(1 point)**

Problem 3 — Making the Most out of Living with a Roommate (26 points)

Imagine a situation in which you're living with a roommate. There are a bunch of household chores that you each have to do each week to be successful adults. Principle among these chores are (1) doing the laundry, and (2) doing the dishes. Unfortunately, you don't have a washer, drier or dish washer, so you have to perform these chores all by hand. What a pain.

You and your roommate each have specific speeds at which you can do each of these tasks. Table 1 below indicates how fast each of you can perform these household chores in a given hour.

Table 1: Productivity in performing household chores (per hour)

	Dishes washed	Clothes washed
You	90	30
Roommate	50	10

- (a) Who has the absolute advantage (you or your roommate) in doing the dishes? **(1 point)**
- (b) Who has the absolute advantage (you or your roommate) in doing the laundry? **(1 point)**
- (c) What is the opportunity cost of washing a dish in terms of clothes washed for **each** you and your roommate. That is, for each you and your roommate, how many clothes do you have to give up washing to wash 1 dish? **(3 points)**
- (d) Who has the higher opportunity cost of washing a dish in terms of clothes washed? That is, who has to give up washing more clothing to wash a dish? **(3 points)**

- (e) What is the opportunity cost of washing an article of clothing in terms of dishes for **each** you and your roommate. That is, for each you and your roommate, how many dishes do you have to give up washing to wash 1 clothing? **(3 points)**
- (f) Who has the higher opportunity cost of washing an article of clothing in terms of dishes? That is, who has to give up washing more dishes to wash 1 clothing? **(3 points)**
- (g) Who has the comparative advantage in washing dishes, and therefore, should specialize in washing dishes? **(1 point)**
- (h) Who has the comparative advantage in doing laundry, and therefore, should specialize in doing laundry? **(1 point)**
- (i) Consider the first week that you and your roommate were living together. Initially, you aren't really feeling your roommate — you're not sure if you like them, or whether they can be

trusted with either your prized Nashville Predators jersey or the specialty kitchen knife you got as a gift from your best friend. And your roommate feels the same about you. So, you each do your own chores.

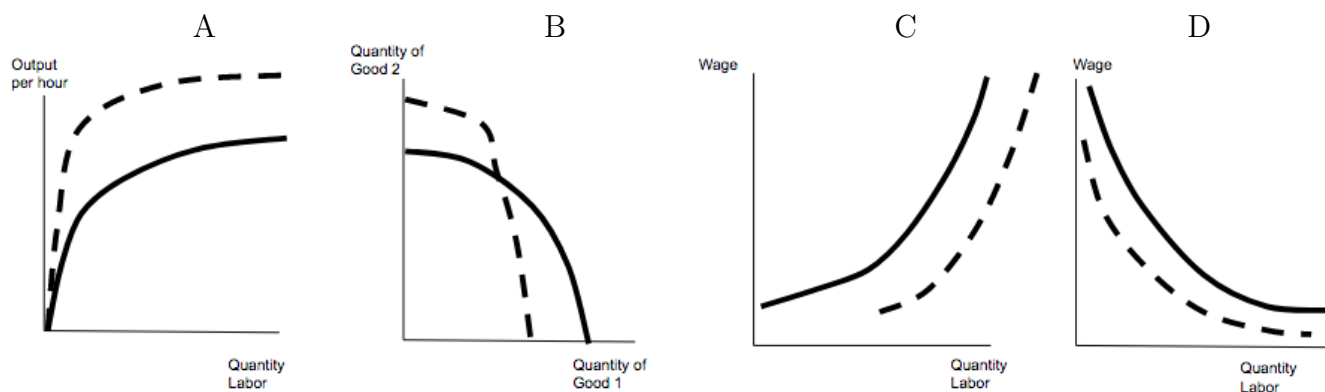
You spend 50 minutes each week washing 75 dishes, and 3 hours (180 minutes) washing 90 pieces of clothing. Your roommate spends 30 minutes each week washing 25 dishes, and 3 hours (180 minutes) washing 30 pieces of laundry. (No judgement that you each have a lot of dishes and clothes!)

But after a week, you and your roommate are much more comfortable with each other and you decide to specialize in the way indicated by your answers to parts (g) and (h) above. If together you and your roommate still need to wash the same number of dishes and pieces of laundry in week 2 as you did in week 1, **how many fewer total minutes (or hours, you can calculate it either way)** will it take you both together to do all of the chores after you've specialized? **(5 points)**

- (j) Consider an economy full of people who undertake specialization between two time periods. In time period 1, each person in the economy is performing all of the productive activities that they need to perform for themselves. In time period 2, each person has specialized in just one productive activity in which they have a comparative advantage and is trading with others for what they themselves do not produce.

In each plot below, time period 1 is indicated by a solid line, and time period 2 is indicated by a dashed line. **Circle the plot (or plots) below** that unambiguously reflect(s) how we can visualize the effect of this specialization in the macroeconomy (all else equal). In just a few sentences at most, **explain briefly why the plot(s) you chose illustrate(s) the effects of specialization, AND why the plot(s) you didn't choose do not illustrate specialization.**

For clarity, plot A is a production function, plot B is a production possibilities frontier, plot C is a labor market supply curve, and plot D is a labor market demand curve. All plots relate to the economy as a whole (not a particular individual, firm, or market). **(5 points)**



Problem 4 — Analyzing Production in a Magical Economy (14 points)

Consider a magical (and therefore clearly fictional) economy. Let's call it Middle Earth. This magical economy produces two products — lumber, and magical wands. Lumber is used as firewood (a final good) to heat homes and cook food because for some reason magic and electricity never go hand in hand. Lumber is also used as an intermediate good (the sole input) in producing magical wands. It takes 2 pounds of lumber to produce 1 magical wand.

Table 1 below provides output and prices for three years in Middle Earth.

Table 2: Production and prices in Middle Earth

Year	Lumber quantity (lbs)	Lumber price (per lb)	Wand quantity (wands)	Wand price (per wand)
Year 1	80	\$1	15	\$10
Year 2	140	\$2	20	\$15
Year 3	124	\$2	22	\$20

- (a) Calculate real GDP for the economy of Middle Earth in each of the three years presented in Table 1 above. If there's a choice you need to make to calculate real GDP, be clear in your answer about what your choice is. **(5 points)**

- (b) Was real GDP growth higher between year 1 and year 2, **or** between year 2 and year 3? **(3 points)**

- (c) Let's say that the population of Middle Earth is 6 people. There is 1 woodcutter who does all of the woodcutting and makes all of his income from cutting wood. There are 5 people who make wands — 1 wand firm owner, and 4 wand-making employees. The firm owner pays each of his employees \$1 for each wand they produce, and keeps the remaining income from selling wands (after he pays for his materials) for himself.

In year 1, the woodcutter buys 5 of the 15 wands produced in Middle Earth. The other 10

wands produced in year 1 are exported to a neighboring magical economy. With their income in year 1, none of the 4 wand employees can afford to buy a new wand. Even in year 2 (after they've worked and been paid), none of the wand employees can afford to buy a new wand. But luckily the woodcutter is willing to sell 4 of his used wands for just \$4 each. Each of the 4 wand-making employees buys one of these used wands in year 2. As a result of this trade, the woodcutter's income has gone up, while the wand-making employees income has gone down. The wand firm owner's income has stayed the same.

What is GDP per capita in year 2 *before* the trade of the used wands, **and** what is GDP per capita *after* the trade? You may calculate real or nominal per capita GDP; whichever you want. (You don't need to consider anything about year 3 when you answer this question.) **(4 points)**

- (d) Answer whether the following statement is **true or false**: If Middle Earth exports all of its magic wands to neighboring Hogwarts each year, its GDP each year is calculated as zero (\$0). In just a few sentences at most, **explain briefly why the statement is true or false.** **(2 points)**